

CONTACT INF

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ABOUT ME

- **Engineer** passionate about understanding how complex systems work.
- **Industrial AI** Applications in Assembly Lines
- **Industry 5.0**

LANGUAGES

- **English:** Advanced
- **French:** Intermediate
- **German:** Basic
- **Spanish:** Native

SKILLS

- **Analytical**
- **Challenge-driven**
- **Decisive**

TECHNICAL SKILLS

- **OS:** Windows, Linux
- **IoT:** MQTT, Openthread, BLE, WebSockets.
- **LANG:** C, C#, Python, Java, HTML, CSS, Kotlin.
- **SIMUL:** Matlab, Simulink
- **CAD:** Catia V5, SolidWorks.
- **AI:** Langgraph, N8N, YOLO.
- **DB:** PostgreSQL, Qdrant.
- **HW:** STM, Esp, Nordic nRF.
- **Tools**: Arduino IDE, VS.
- **IND:** Opc UA, Scada, PLC

WHAT I HAVE BUILT

MY PROJECTS

Francisco Sierra González

Electrical, Electronics & Industrial AI Engineer

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PROFESSIONAL EXPERIENCE

Systems Engineer — CT Engineering — 2026-present

IoT Department/AI Aeronautical Systems

- **Airbus Hangar Centralization & Multi-Agent AI (ZAL):** Engineered an IoT platform with interactive maps and heatmaps. Built a multi-agent backend (Orchestrator, Chatbot, and Frame Analyzer) to autonomously onboard and parse new sensors.
- **OpenThread Mesh Sensor Network & Gateway:** Deployed a scalable ESP32/Nordic nRF environmental mesh network with a custom CLI IoT Gateway and MQTT telemetry.
- **AI Connector Inspection System (Airbus):** Developed an automated QA solution for aircraft wire harnesses using custom YOLO for real-time pin detection and 3D point-cloud processing against XML specs.
- **Smart Freezer Automation (PLC & LoRa):** Designed PLC control, power management, and long-range LoRa wireless telemetry for industrial freezer units.

AI Research Associate — Univ. of Seville — 2026-present

Industrial Design Engineering Department

- **AI-driven Operational Excellence System.** Integrates artificial intelligence directly into traditional continuous improvement frameworks like Lean, Six Sigma..
- **Cognitive Industrial Machines based on Industry 5.0 for Aircraft Assembly Lines.** Developed applications using Generative AI (Embodied AI) to provide cognitive capabilities to anthropomorphic industrial systems.

Intelligent Product Development Engineer (Intern) — Univ. of Seville — 2025

Industrial Design Engineering Department

- **Transformed King Air 360 Aircraft Functional Test Areas** by developing a RAG (Retrieval-Augmented Generation) multi-agent architecture to assist operators during test execution.
- **Aircraft Final Assembly Line Transformation through AI for Industry 5.0.** Study of a fictional FAL under Industry 4.0 (Boeing, Airbus...) leveraging Generative AI potential to transform it into a sustainable, resilient, and human-centric Industry 5.0.
- **Designed and established the new CATEPS Laboratory.**.

EDUCATION

Artificial Intelligence

Specialist in Artificial Intelligence Applications	Internat. Univ. Andalucía	2025
Deep Research with LangGraph	LangGraph	2026
Complete Python Course	Udemy	2025
Artificial Intelligence. Language Models	Internat. Univ. Andalucía	2024

Electronics

Electronics Engineering (Double Degree)	Univ. of Seville	2020 - 2026
Cobot UR20/30 Universal Robots	UR Academy	2024
Cobots Risk Assessment e-Learning	UR Academy	2024
Sensorized Industrial Plant Project	Univ. of Seville	2022

Electrical

Electrical Engineering (Double Degree)	Univ. of Sevilla	2020 - 2026
Aircraft Electrical Systems	Udemy	2026

Manufacturing

Lean Six Sigma Green Belt	Udemy	2025
Aircraft Functional Tests Course	Airbus	2023